MIL-R-45600A(Ord)
26 October 1959
SUPERSEDING
MIL-R-45600 (Ord)
31 December 1958

MILITARY SPECIFICATION

REGULATOR, VOLTAGE, CN-290/M

1. SCOPE

1.1 This specification covers one type of voltage regulator which is used in the fire control and guided missile systems listed in 6.1.1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-202	Test Methods for Electronics and Electrical Components Parts

DRAWINGS

Ordnance Corps

D7614318	Regulator, Voltage, CN-290/M, Schematic Diagram
D8175023	Connector, ~~Receptable~~ Receptacle, Electrical
E7620535	Regulator, Voltage, CN-290/M

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one regulator, representative of production, shall be made available to the Government to determine if the regulator meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 Design and Construction.- The regulator shall be manufactured in accordance with Drawing E7620535, "Regulator, Voltage, CN-290/M" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance, and permanence capabilities prior to final acceptance by the Government.

3.4 Ground potential.- The design and construction of the regulator shall insure that all external parts are at ground potential at all times.

3.5 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Electron tubes
(f) Workmanship

3.6 Environmental.

3.6.1 Vibration.- The regulator shall be subjected to a simple harmonic motion having an amplitude of 0.03-inch (0.06-inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second return to 10 cycles and shall be traversed in approximate 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration the regulator shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.6.2 Operating temperatures.- The regulator shall meet all the performance requirements of this specification when thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit.

3.6.3 Storage temperatures.- The regulator shall show no evidence of physical failure and shall meet all performance requirements of this specification at standard ambient temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

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3.7 Electrical.

3.7.1 Continuity.- The regulator shall demonstrate continuity in accordance with Electrical Schematic Drawing D7614318.

3.7.2 Dielectric strength.- There shall be no puncture of insulation, no spark-over, or decrease in dielectric strength when the insulated areas of the regulator are subjected to an alternating-current (a-c) root mean square (rms) potential of 1000 volts plus twice the rated voltage at a frequency of 60 cycles per second (cps) for a period of not more than 30 seconds nor less than 5 seconds.

3.7.3 Insulation resistance.- Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current applied.

3.8 Performance.- The regulator, when set-up in accordance with pretest information specified in Section 4, shall demonstrate the following requirements. All symbols referred to in the following paragraphs are those shown on Drawing D7614318.

3.8.1 Output voltage.- The output voltage of the regulator shall be minus 200 plus or minus 4 volts.

3.8.2 Output noise.- The output noise shall be less than 2 millivolts (mv).

3.8.3 Regulating range.- The output current shall be capable of being varied from 440 to 740 milliamperes (ma). Through this range, the output voltage shall remain less than 2 mv.

3.8.4 Output impedance.- The output voltage shall be between 2.5 and 4.0 mv at a frequency of 800 cycles per second (cps) between 8.5 and 13.5 mv at 8000 cps; and between 14.0 and 31.0 mv at 80,000 cps.

3.8.5 Balance.- The balance output voltages shall be within 10 volts of one another.

3.9 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Contractor's quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

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(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

4.1.1.1 The contractor will not be restricted to the inspection station or to the method of inspection listed, provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine whether the practices, methods and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government will report such instances.

4.2 Classification of tests.- The inspection and testing of the regulator shall be classified as follows:

(a) Production model test
(b) Acceptance test

4.2.1 Production model.

4.2.1.1 Unless otherwise specified in the contract, one unit taken from the production run shall be tested by the contractor under the surveillance of the Government.

4.2.2 These tests shall be conducted at the contractor's facilities using test equipment designed or procured for use on the remaining items.

4.2.3 The model shall meet all the requirements of this specification using the inspection methods prescribed herein.

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4.2.3.1 Disposition of rejected production model.- Should the production model fail to pass the test for Government acceptance, production of those components causing rejection will cease until the defective characteristics have been corrected in the production process. When corrections in the process have been made, a new production model shall be completely inspected by the Government. The rejected production model shall be corrected, where practical, and shall become part of the first production lot.

4.3 Inspection provisions.

4.3.1 Inspection of components.- All positional and functional characteristics shown on the drawings contained in the contract shall be included in the inspection plan and shall be inspected by the contractor for conformance with drawing requirements prior to acceptance. These characteristics are defined as all dimensions and functional notes on which the tolerance indicates a critical requirement having a bearing on the functional accuracy and performance of the regulator.

4.3.1.1 Sampling inspection.- Sampling inspection of components by the contractor may be used in lieu of 100 percent inspection and shall be in accordance with Table I.

TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - over	30	1	2

4.3.1.1.1 Disposition of rejected lots.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. All rejected lots shall be screened for all defective characteristics. After corrections have been made and the government inspector informed of the corrected action taken, the lot shall be resubmitted for acceptance inspection of the characteristics that caused the lot to be rejected.

4.3.1.2 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.3.2 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

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4.3.3 Acceptance tests.- The classification of defects in tables II and III shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items. Samples selected for acceptance test shall include the sample selected for environmental testing.

4.3.3.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE II - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

For Reduced Inspection use Level I, and Sampling Plan Table IV-A of Standard MIL-STD-105

AQL values are expressed in percent defective.

MAJOR: AQL 1.5%	Req't Par.	QA Par.	Test Method
1. Continuity	3.7.1	4.4.1	Standard Measuring Equipment (SME)
2. Dielectric strength	3.7.2	4.4.2	SME
3. Insulation resistance	3.7.3	4.4.3	SME
4. Output voltage	3.8.1	4.4.4	SME
5. Output noise	3.8.2	4.4.5	SME
6. Regulating range	3.8.3	4.4.6	SME
7. Output impedance	3.8.4	4.4.7	SME
8. Balance	3.8.5	4.4.8	SME
9. Identification-marking	3.9	4.4.12	Visual

TABLE III - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level L6, and Sampling Plan Table IV-A of Standard MIL-STD-105

MAJOR: AQL 4.0%	Req't Par.	QA Par.	Test Method
1. Vibration	3.6.1	4.4.9	DST
2. Operating temperatures	3.6.2	4.4.10	DST
3. Storage temperatures	3.6.3	4.4.11	DST

4.3.4 Inspection equipment.

4.3.4.1 Contractor provisions.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also

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shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.3.4.2 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.3.4.2.1 In addition to the above, the test equipment listed below shall be connected as shown on Figure 1, and as described herein.

Item No.	Description
1	Electron tube voltmeter, frequency range 10 to 15,000 cps, voltage range 0.001 volt to 100 volt rms, input impedance 0.5 megohms, shunted by 30 micromicrofarad (uuf), accuracy 2 percent over the entire range.
2	Oscillator, capable of providing a frequency range of 5 cps to 600 oscillator kilocycle (kc), frequency response plus or minus 1 decibel (db), internal impedance 600 ohms, distortion less than 1 percent, frequency may be standardized with an accuracy of 1 percent.
3.	Voltmeter, dc. 0-500 volts, 1000 ohms per volt plus or minus 2 percent.
4.	Ammeter, 0-800 milliamperes, plus or minus 2 percent.
5.	Resistor, 3000 ohms plus or minus 10 percent, 250 watts.
6.	Rheostat, 1000 ohms plus or minus 5 percent, 250 watts.
7.	Resistor, 500 ohms, plus or minus 5 percent, 1 watt.
8.	Resistor, 1 megohm, plus or minus 20 percent, 1/2 watt.
9.	Capacitor, 1 microfarad, plus or minus 10 percent, 600 volts.
10.	Switch, single pole, double throw.
11.	Switch, 1 circuit, 5 position.
12.	Connector, per Ordnance Corps Drawing D8175023.
13.	Capacitor, 0.02 microfarad plus or minus 10 percent, 600 volts.

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Item No.	Description
14.	Resistor, 200 ohms plus or minus 20 percent, 1/2 watt.
15.	Capacitor, 450 volts, 10 microfarads, plus or minus 10 percent.
16.	Power supply, (6.3 volts, 3.0 amperes, 400 cycles, single phase).
17.	Power supply, (minus 250 volts, 1.0 milliamperes, regulated plus or minus 2 volts less than 10 mv hum)
18.	Power supply (minus 320 volts, 0.790 ampere, regulated plus or minus 3 volts).

4.3.4.3 Rheostat, item 6, shall be adjusted, after the filaments have been heated and the d-c power applied, until meter, item 4, reads 590 plus or minus 10 ma. The minus 320 volt input, item 18 shall be adjusted to minus 320 plus or minus 2 volts as read on meter, item 3, with switch, item 11, in position 1. The minus 250 volt input, item 17, shall be adjusted to minus 250 plus or minus 2 volts as read on meter, item 3, with switch, item 11, in position 2.

4.4 Test methods and procedures.

4.4.1 Continuity.- A continuity check of the regulator circuits shall be made using standard measuring equipment to determine continuity with the applicable drawing as specified in 3.7.1.

4.4.2 Dielectric strength.- The regulator shall be tested in accordance with method 301 of Standard MIL-STD-202, and shall meet the requirements specified in 3.7.2.

4.4.3 Insulation resistance.- Immediately after satisfactory withstanding of the dielectric strength test, the insulation resistance test shall be performed on the unit under test in accordance with method 302, Test Condition "B", of Standard MIL-STD-202, and shall meet the requirements of 3.7.3.

4.4.4 Output voltage.- With switch, item 11, in position 3; meter, item 3, shall indicate the output voltage specified in 3.8.1.

4.4.5 Output noise.- With meter, item 1, connected to the J3 terminals, the reading shall be as specified in 3.8.2.

4.4.6 Regulating range.- With switch, item 11, in position 3; switch, item 10, in position 1, and by adjusting rheostat, item 6; the reading of meter, item 4, shall indicate the output current as specified in 3.8.3 and the output noise voltage, when read on meter, item 1, shall be as specified in 3.8.3. The reading on meter, item 4 shall be returned to 590 plus or minus 10 ma.

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4.4.7 Output impedance.

(a) Oscillator, item 2, shall be connected to the J4 terminals.

(b) With switch 10 in the 2 position and oscillator, item 2, at 800 cps the output of the oscillator, item 2, shall be adjusted until meter, item 1, reads 5 volts. With switch, item 10, switches to position 1, item 1, shall read as specified in 3.8.4.

(c) The procedure outlined in 4.4.7(b) shall be repeated at 8000 cps. With Switch, item 10, switched to position 1, the meter, item 1, shall read as specified in 3.8.4.

(d) The procedure outlined in 4.4.7(b) shall be repeated at 80,000 cps. With switch 10 in position 1, meter, item 1, shall read as specified in 3.8.4.

4.4.8 Balance.- The readings of meter, item 3, with switch, item 11, in positions 4 and 5 shall be as specified in 3.8.5.

4.4.9 Vibration.- The regulator shall be rigidly mounted on the vibration table in its normal mounting operating position, and it shall be vibrated successively along each of the three mutually perpendicular directions that are respectively parallel to the edge of the regulator with an amplitude, duration, and frequency as specified in 3.6.1. At the conclusion of this test, the voltage regulator shall be visually inspected for physical damage, then tested in accordance with 4.4.4 to 4.4.8 inclusive. The measuring instruments for checking the vibration amplitude shall be capable of measuring within plus or minus 5 percent and capable of measuring the vibration frequency within plus or minus 2 percent.

4.4.10 Operating temperatures.- The regulator shall be placed in a test chamber capable of the temperature requirements of 3.6.2. The test shall be in accordance with Procedure I of Specification MIL-F-13926(Ord) and then shall meet the performance requirements specified.

4.4.11 Storage temperatures.- The regulator shall be placed in a test chamber capable of the temperature requirements of 3.6.3. The test shall be in accordance with Procedure II of Specification MIL-F-13926(Ord). After exposure, the regulator shall be visually examined for evidence of physical failure and then shall meet the requirements specified in 3.6.3.

4.4.12 Identification marking.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

4.4.13 Packaging, packing and marking.- Quality Assurance Provisions of packing, packaging, and marking shall be in accordance with Specification MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- The regulator shall be cleaned, preserved, packaged, packed and marked in accordance with Specification MIL-F-14619 P81. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use.- The regulator is a shunt type regulator intended to supply minus 200 volts at 590 milliamperes to the cathodes of the power stages of the direct-current amplifiers.

6.1.1 Application.- The regulator is a component of the following assemblies and systems:

Assembly	System
Computer	Antiaircraft Fire Control System M33C (90mm) and M33D (120mm)
Power Supply Group OA-742/M	Radar Course Directing Central AN/MSW-1 and AN/MSW-3
Power Supply Group OA-1527/MSA-19	Antiaircraft Guided Missile System-Nike Hercules

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1).

6.3 Definition.

6.3.1 Standard measuring equipment (SME).- Standard measuring equipment (SME) includes the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

6.3.2 Device, special testing (DST).- Special test devices are designed to check function of an assembled unit.

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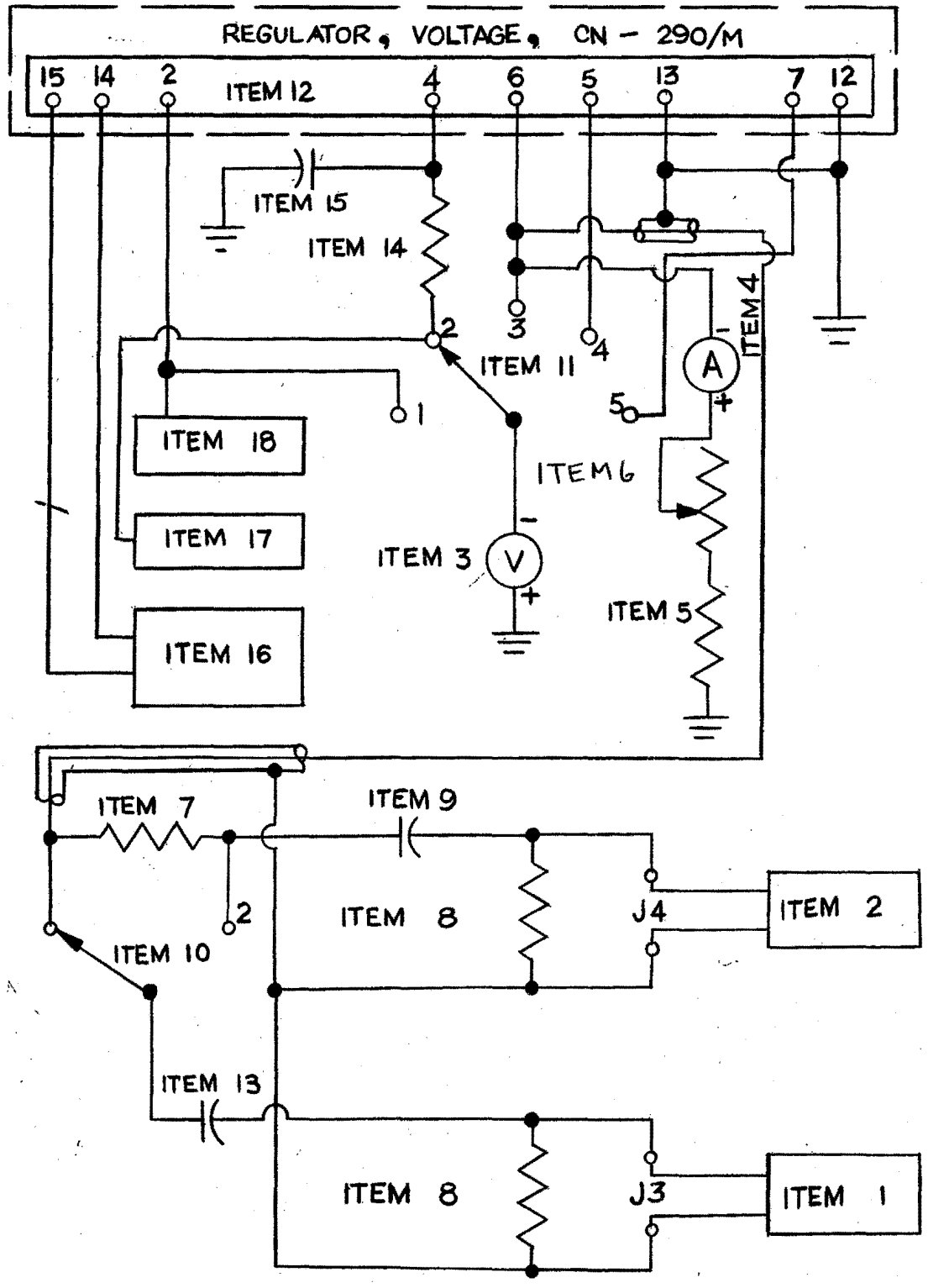


FIGURE 1

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